

PisoFire M1

**High-Demand Sneaker
E-commerce Platform**

Group D

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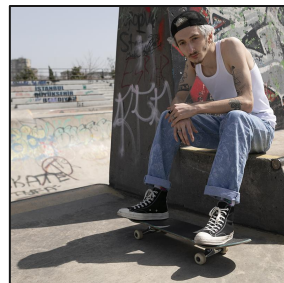
What is PisoFire?

- **Product** : Cloud-native e-commerce platform dedicated to premium sneaker releases.
- **Industry Domain** : High-volume retail focusing on "Flash Sales" .
- **SRE Challenge**: The system must operate reliably during periods of constant traffic; maintain availability and data integrity during “Flash Sales”.

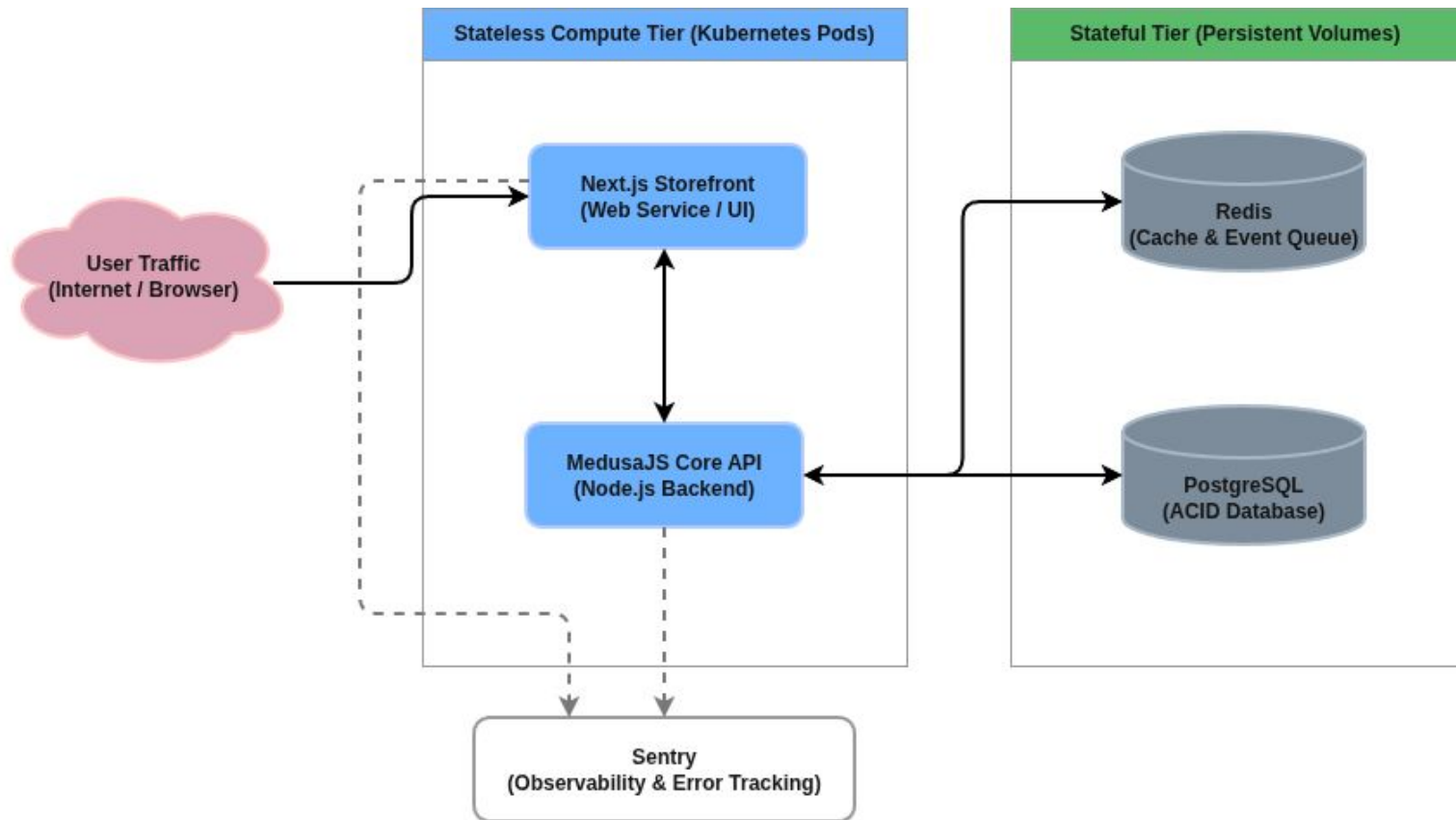
Critical Path

Flash Sales Buying Path

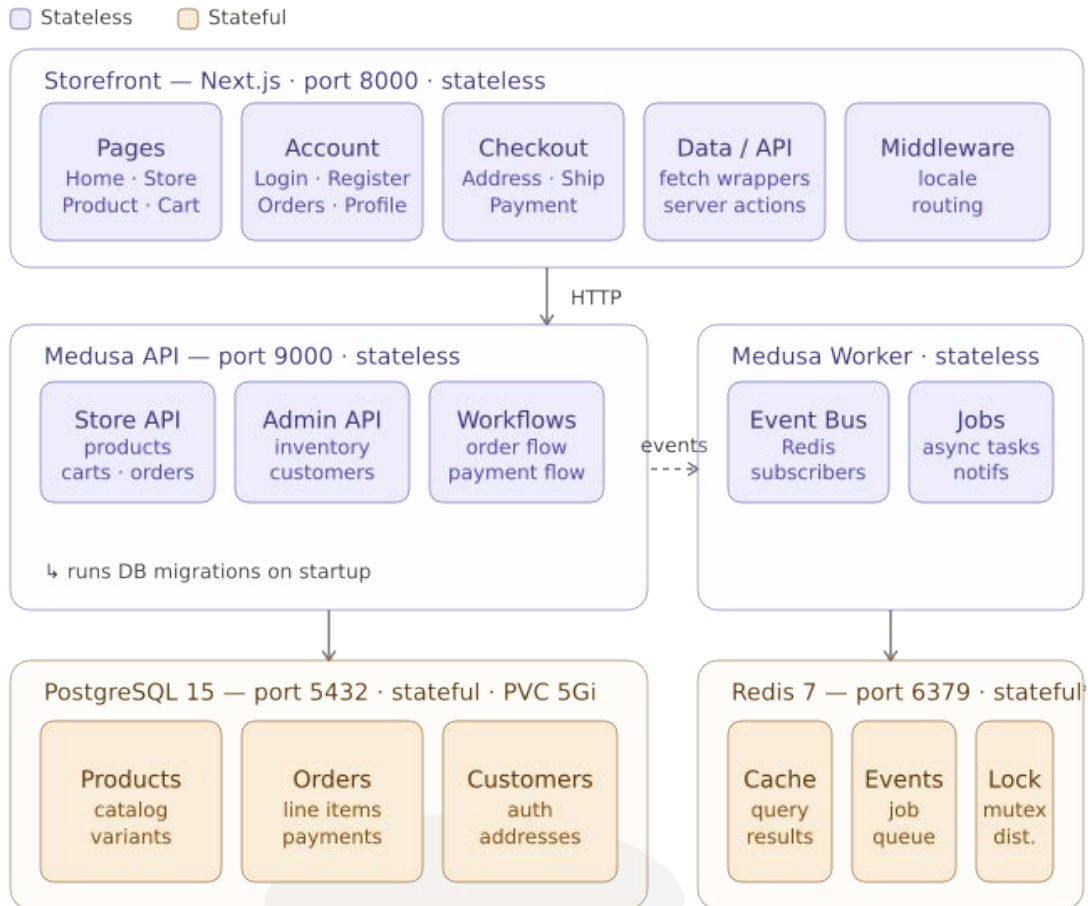
- User goes to flash sale landing page after launching time
- User can view the product's details
- User selects a size
- User adds item to cart
- User goes to checkout
- Payment is processed
- Inventory is decreased



M0 Architecture



M1 Architecture



Strategy

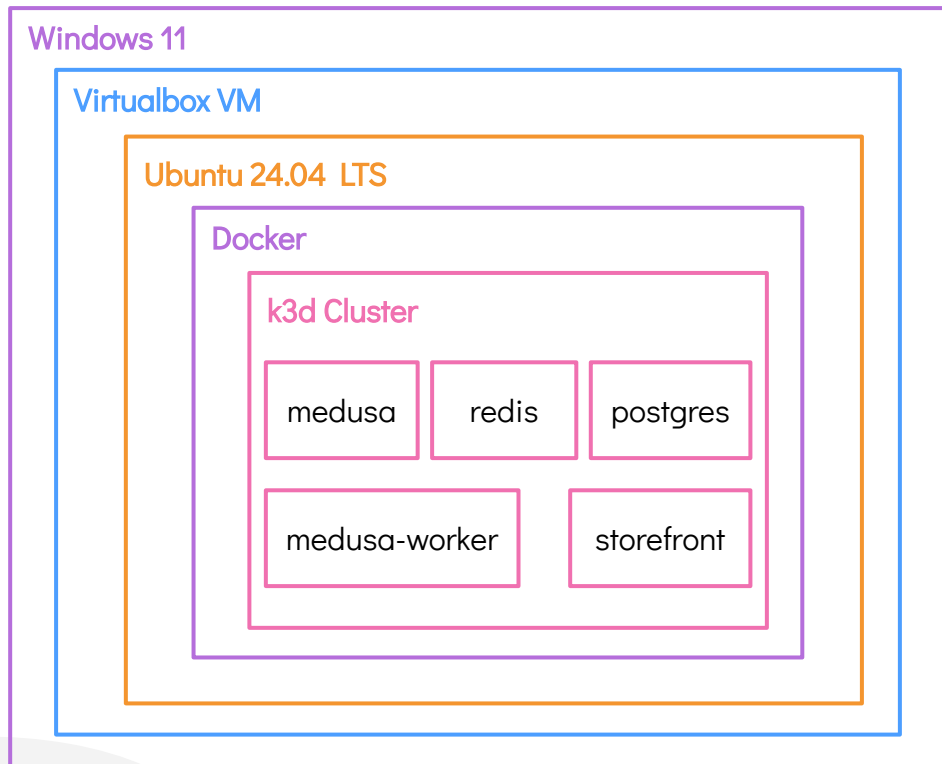
- Local Kubernetes cluster
- Docker images imported into the k3d cluster
- Kubernetes manifests managed with Kustomize
- Deployment automated with one script

```
./scripts/deploy-k8s.sh
```



Deployment Environment

- Local Ubuntu 24.04 LTS VM
- Running on VirtualBox
- Hosted on a Windows 11 machine
- k3d Kubernetes cluster
- Docker as the container



Kubernetes Component Mapping

Product Component	Kubernetes resources
Frontend	Deployment, Service
Backend	Deployment, Service
Backend worker	Deployment
PostgreSQL	Deployment, Service, PVC
Redis	Deployment, Service
Store Bootstrap	Job

NAME	READY	STATUS	RESTARTS	AGE
pod/bootstrap-store-4gfvd	0/1	Completed	0	175m
pod/medusa-756c5c5bcc-ndk8v	1/1	Running	1 (4m19s ago)	175m
pod/medusa-worker-b47bbb579-mcqwz	1/1	Running	2 (4m19s ago)	175m
pod/postgres-799f9b5579-9xs8h	1/1	Running	1 (4m19s ago)	175m
pod/redis-7947c4c48b-wrnmf	1/1	Running	1 (4m19s ago)	175m
pod/storefront-67c95d8b5d-j965j	1/1	Running	1 (4m19s ago)	175m

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/medusa	ClusterIP	10.43.1.223	<none>	9000/TCP	175m
service/postgres	ClusterIP	10.43.241.62	<none>	5432/TCP	175m
service/redis	ClusterIP	10.43.142.129	<none>	6379/TCP	175m
service/storefront	ClusterIP	10.43.188.190	<none>	8000/TCP	175m

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/medusa	1/1	1	1	175m
deployment.apps/medusa-worker	1/1	1	1	175m
deployment.apps/postgres	1/1	1	1	175m
deployment.apps/redis	1/1	1	1	175m
deployment.apps/storefront	1/1	1	1	175m

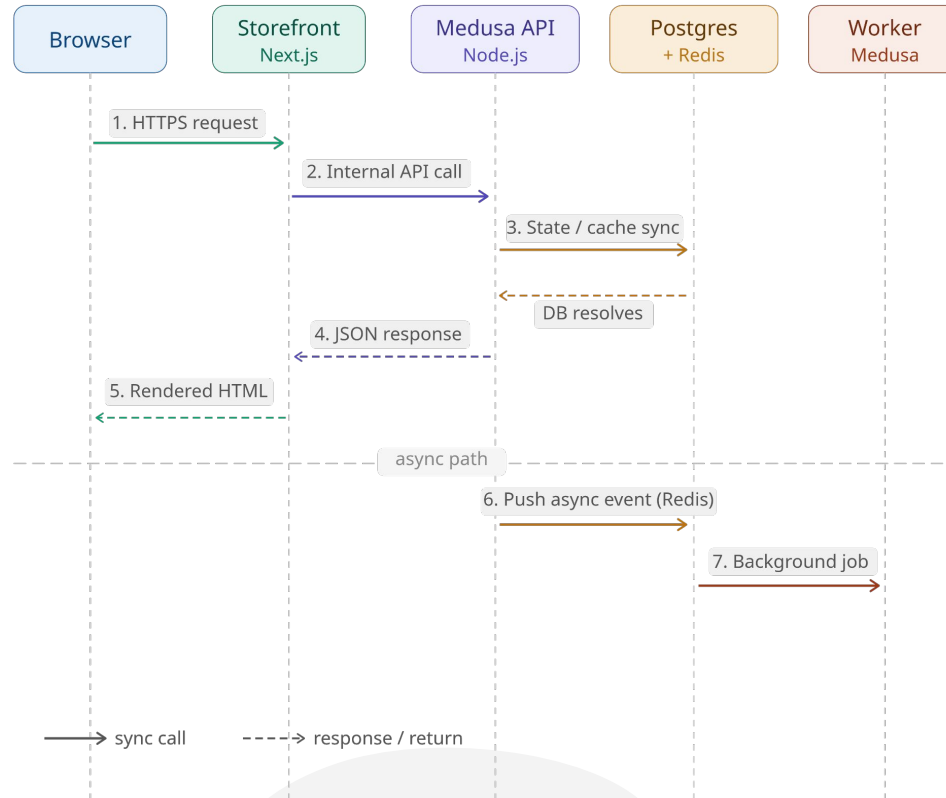
NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/medusa-756c5c5bcc	1	1	1	175m
replicaset.apps/medusa-worker-b47bbb579	1	1	1	175m
replicaset.apps/postgres-799f9b5579	1	1	1	175m
replicaset.apps/redis-7947c4c48b	1	1	1	175m
replicaset.apps/storefront-67c95d8b5d	1	1	1	175m

NAME	STATUS	COMPLETIONS	DURATION	AGE
job.batch/bootstrap-store	Complete	1/1	60s	175m

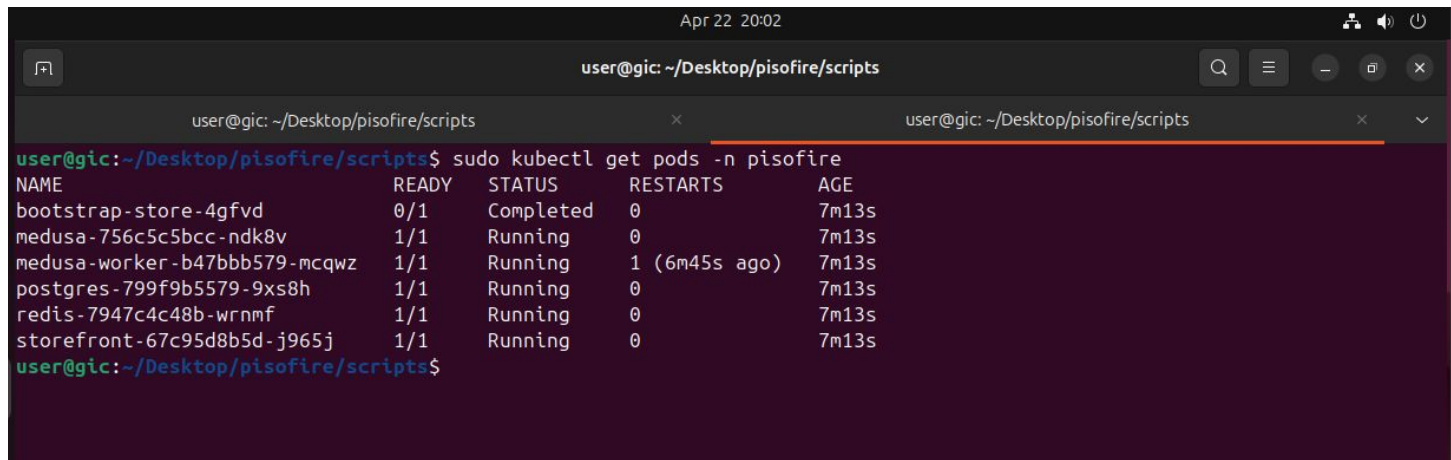
Health Checks

Component	Probe Method	Checks
Postgres	pg_isready	Readiness + Liveness
Redis	TCP port	Readiness + Liveness
Medusa API	/health	Readiness + Liveness
Medusa Worker	/health	Readiness + Liveness
Storefront	/api/health	Startup + Readiness + Liveness

Data Flow



Pods

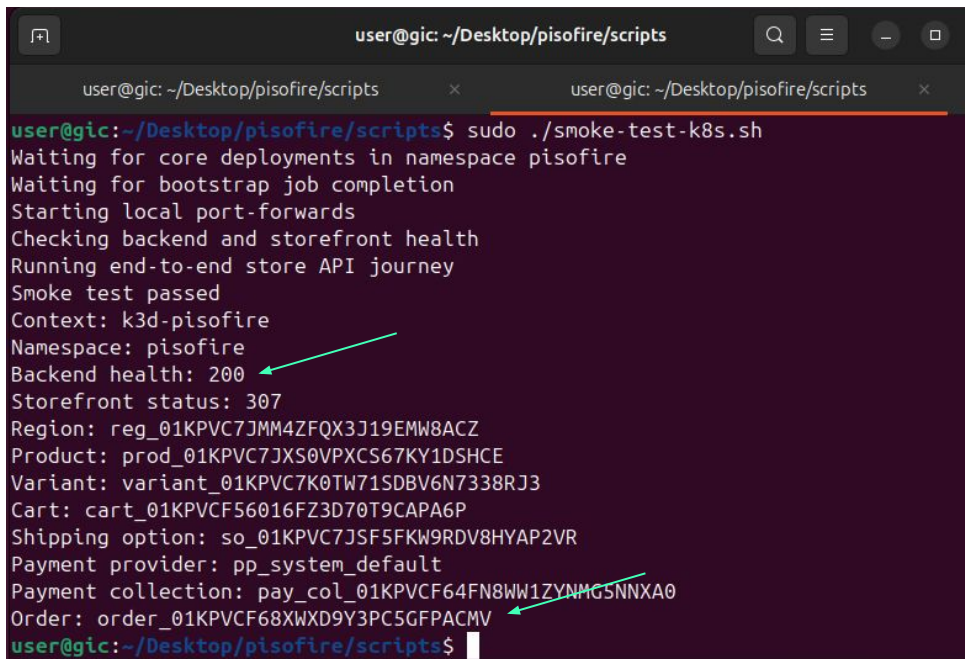
A terminal window with a dark background and light text. The title bar at the top shows the date and time 'Apr 22 20:02' on the right and the window title 'user@gic: ~/Desktop/pisofire/scripts' in the center. Below the title bar, there are two tabs, both with the same title. The terminal content shows a command being executed: 'sudo kubectl get pods -n pisofire'. The output is a table with five columns: NAME, READY, STATUS, RESTARTS, and AGE. The rows list various pods, including 'bootstrap-store-4gfvd' which is 'Completed', and several others in a 'Running' state. The prompt 'user@gic:~/Desktop/pisofire/scripts\$' is visible at the bottom.

```
Apr 22 20:02
user@gic: ~/Desktop/pisofire/scripts
user@gic: ~/Desktop/pisofire/scripts
user@gic:~/Desktop/pisofire/scripts$ sudo kubectl get pods -n pisofire
NAME                                READY   STATUS    RESTARTS   AGE
bootstrap-store-4gfvd               0/1     Completed 0           7m13s
medusa-756c5c5bcc-ndk8v             1/1     Running   0           7m13s
medusa-worker-b47bbb579-mcq wz      1/1     Running   1 (6m45s ago) 7m13s
postgres-799f9b5579-9xs8h          1/1     Running   0           7m13s
redis-7947c4c48b-wrnmf              1/1     Running   0           7m13s
storefront-67c95d8b5d-j965j         1/1     Running   0           7m13s
user@gic:~/Desktop/pisofire/scripts$
```

Smoke Tests

The smoke test:

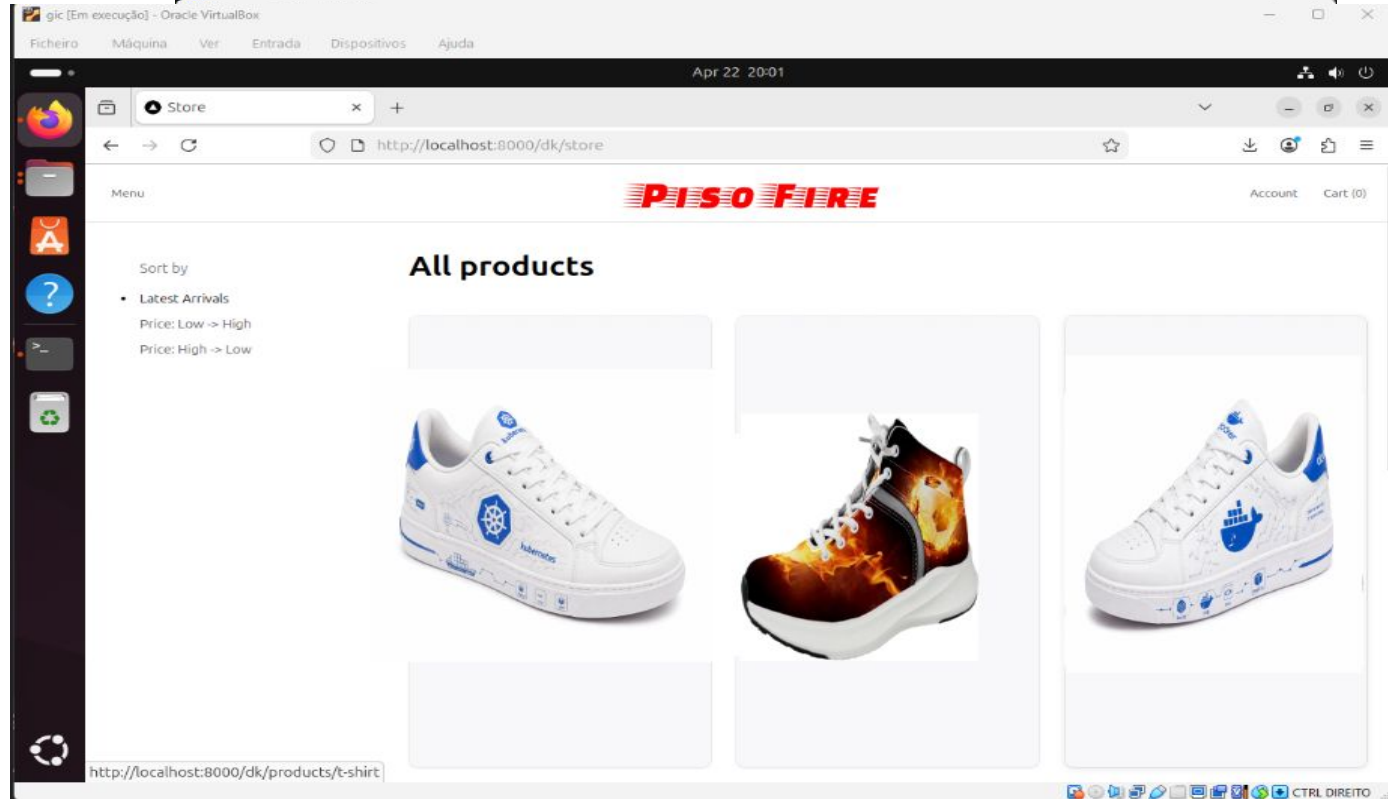
- Waits for Pods and Deployments
- Checks health endpoints
- Validates Store API flow
- Creates cart and order



```
user@gic: ~/Desktop/pisofire/scripts
user@gic: ~/Desktop/pisofire/scripts x user@gic: ~/Desktop/pisofire/scripts x
user@gic:~/Desktop/pisofire/scripts$ sudo ./smoke-test-k8s.sh
Waiting for core deployments in namespace pisofire
Waiting for bootstrap job completion
Starting local port-forwards
Checking backend and storefront health
Running end-to-end store API journey
Smoke test passed
Context: k3d-pisofire
Namespace: pisofire
Backend health: 200
Storefront status: 307
Region: reg_01KPVC7JMM4ZFQX3J19EMW8ACZ
Product: prod_01KPVC7JXS0VPXCS67KY1DSHCE
Variant: variant_01KPVC7K0TW71SDBV6N7338RJ3
Cart: cart_01KPVCF56016FZ3D70T9CAPA6P
Shipping option: so_01KPVC7JSF5FKW9RDV8HYAP2VR
Payment provider: pp_system_default
Payment collection: pay_col_01KPVCF64FN8WW1ZYNNMG5NNXA0
Order: order_01KPVCF68XWD9Y3PC5GFPACMV
user@gic:~/Desktop/pisofire/scripts$
```

The image shows a terminal window with a dark background. The prompt is 'user@gic:~/Desktop/pisofire/scripts'. The command executed is 'sudo ./smoke-test-k8s.sh'. The output shows the test steps: waiting for deployments, starting port-forwards, checking health, and running the API journey. It reports 'Smoke test passed' and lists various context and namespace details. Two green arrows point to 'Namespace: pisofire' and 'Payment collection: pay_col_01KPVCF64FN8WW1ZYNNMG5NNXA0'.

Our XUXOP (Shoe Shop)



What Can Go Wrong?

- Single replica per component
- PostgreSQL is a single point of failure
- Redis is short-lived and single-instance
- No autoscaling
- Limited observability



Thanks for your attention!

Any feedback among us?

